

Brian Lai Lap Hong

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Education

Universiti Teknologi Malaysia (UTM) Master of Philosophy (Robotics & Localization)	2022–2025
Universiti Teknologi Malaysia (UTM) Bachelor of Electronic Systems Engineering	2017–2021

Profile

Robotics engineer specializing in C++, Python, ROS1/ROS2, sensor fusion, and autonomous systems. Experienced in simulation, machine learning model implementation, and deploying solutions on Linux-based platforms. Adept at full-cycle project delivery from simulation to hardware deployment, collaborating effectively with multidisciplinary teams.

Work Experience

Robotics Software Engineer, Braintree Technologies	2021–Present
- Designed and implemented autonomous navigation systems using ROS1/ROS2 and C++ for agricultural mobile robots and field-based AGVs. - Developed robotic solutions for precision agriculture applications, including fertilizer deployment, crop monitoring, and oil palm plantation navigation. - Built localization and navigation pipelines for outdoor, unstructured environments, handling uneven terrain, vegetation, and GPS variability. - Applied machine learning for crop, obstacle, and human detection to enable safe and environment-aware operation. - Developed simulation workflows in Gazebo to model field environments, followed by validation through real-world deployment. - Processed and fused sensor data (LiDAR, cameras, IMU, GPS) for robust SLAM and localization in dynamic outdoor conditions. - Deployed systems on embedded platforms (e.g., Jetson) for real-time onboard processing in agricultural robots. - Integrated perception and data pipelines to support data-driven farming and operational insights. - Collaborated with vendors and hardware suppliers to integrate sensors, embedded systems, and robotic components for deployment. - Worked closely with customers and field operators to gather requirements, conduct on-site testing, and refine robotic solutions based on real-world feedback. - Coordinated with cross-functional engineering teams to deliver end-to-end agricultural automation systems.	

Postgraduate Experience

MPhil Researcher, Universiti Teknologi Malaysia	2022–Present
- Conducting advanced research on localization techniques for mobile robotics. - Enhanced scan-matching algorithms to improve positional accuracy and robustness. - Built agricultural simulation environments to benchmark and compare localization methods. - Performed literature reviews of state-of-the-art techniques to inform research direction.	

Technical Skills

Programming: C++, Python, MATLAB

Robotics & Simulation: ROS1, ROS2, Gazebo, Blender, localization, mapping, navigation stacks

Hardware & Systems: IMU, RealSense, Jetson, Arduino, Teensy, sensor fusion, PCB prototyping, Linux, networking, Git

Software & Tools

Photoshop, Blender, Premiere Pro

Languages

Cantonese (Native), Mandarin (Native), Malay (Fluent), English (Fluent)

Publication

Brian Lai Lap Hong; Mohd Azri Bin Mohd Izhar; Norulhusna Binti Ahmad. 2025

Improved Monte Carlo Localization for Agricultural Mobile Robots with the Normal Distributions Transform.

International Journal of Advanced Computer Science and Applications (IJACSA), 16(3), 2025

DOI: [10.14569/IJACSA.2025.01603100](https://doi.org/10.14569/IJACSA.2025.01603100) [↗](#)

Honors & Competitions

RoboCup Asia-Pacific (RCAP) 2019, Moscow, Russia 2019

- Served as Team Leader of the UTM robotics team in an international robotics competition.
- Led system architecture design and technical decision-making across software and hardware modules.
- Coordinated development of autonomous robot functionalities, including navigation and perception.
- Achieved strong competition results (featured in official UTM news).
- Article: [UTM News](#) [↗](#)

RoboCup Asia-Pacific (RCAP) 2018 2018

- Served as Team Leader of Team Innokai, representing UTM in an international robotics competition.
- Led robot system integration and contributed to navigation algorithm development.
- Guided the team to achieve **2nd Place** in the competition.
- Article: [UTM News](#) [↗](#)

Reference

Dr. Mohd Azri bin Mohd Izhar – Lecturer, Universiti Teknologi Malaysia; Email: mohdazri.kl@utm.my [↗](#)